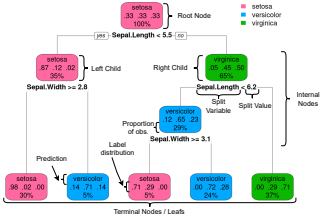
# Introduction to Machine Learning

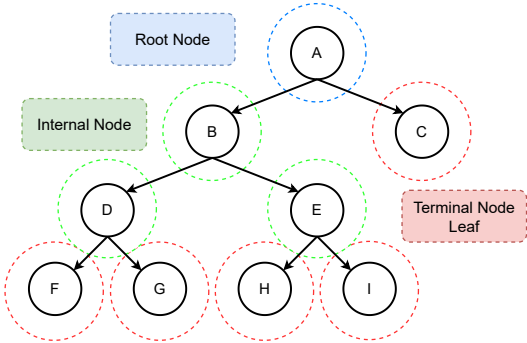
# CART
# Predictions with CART




## Learning goals

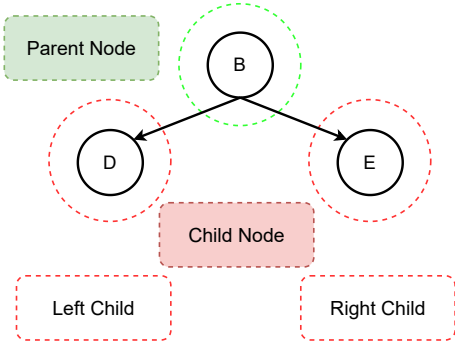
- Understand the basic structure of a tree model
- Understand that the basic idea of a tree model is the same for classification and regression
- Understand how the label of a new observation is predicted via CART
- Know hypothesis space of CART

# BINARY TREES



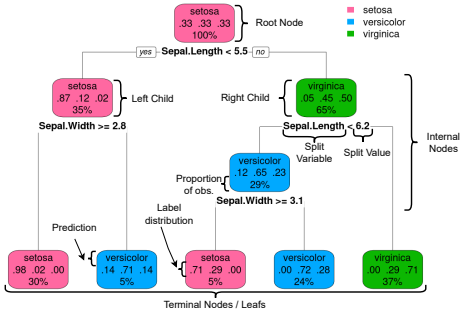

- Binary trees represent a top-down hierarchy with binary splits
- A tree is composed of different node types: a) the root node, b) internal nodes, and c) terminal nodes (also leaves).

# BINARY TREES




- Nodes have relative relationships, they can be:
  - Parent nodes
  - Child nodes
- Root nodes don't have parents – leaves don't have children

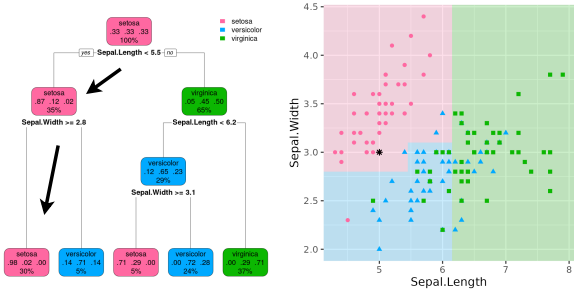
# CLASSIFICATION TREES




- Classification trees use the structure of a binary tree
- Binary splits are constructed top-down in a *data optimal* way
- Each split is a threshold decision for a single feature
- Each node contains the training points which follow its path
- Each leaf contains a constant prediction

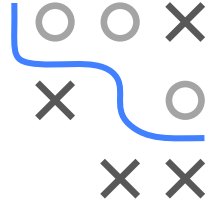
# CLASSIFICATION TREE MODEL AND PREDICTION

- When predicting new data (here*: `Sepal.Length` = 5, `Sepal.Width` = 3) we use the learned split points and pass an observation through the tree
- Each observation is assigned to exactly one leaf
- Classification trees can make hard-label predictions (here: `setosa`) or predict probabilities (here: 0.98, 0.02, 0.00)

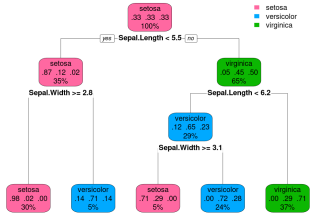


# CART AS A RULE BASED MODEL

Leaf nodes can be expressed by a set of rules (left to right):

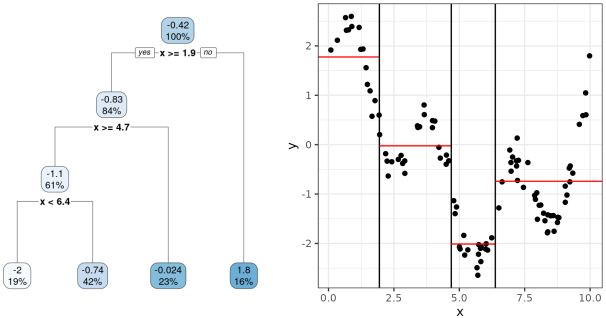


| Hard label prediction | Label distribution | `Sepal.Width` | `Sepal.Length` |
|---|---|---|---|
| `setosa` | 0.98, 0.02, 0.00 | $\geq 2.8$ | $< 5.5$ |
| `versicolor` | 0.14, 0.71, 0.14 | $< 2.8$ | $< 5.5$ |
| `setosa` | 0.71, 0.29, 0.00 | $\geq 3.1$ | $\geq 5.5$ & $< 6.2$ |
| `versicolor` | 0.00, 0.72, 0.28 | $< 3.1$ | $\geq 5.5$ & $< 6.2$ |
| `virginica` | 0.00, 0.29, 0.71 | – | $\geq 6.2$ |



# REGRESSION TREE MODEL AND PREDICTION

- Works the same way as for classification
- But predictions in leaf nodes are a numerical scalar

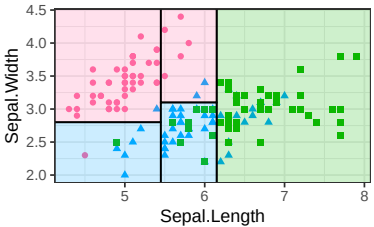


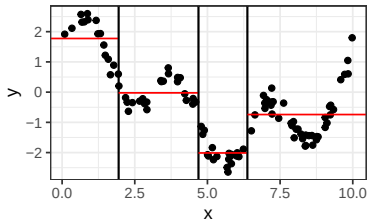
# TREE AS AN ADDITIVE MODEL

Trees divide the feature space $\mathcal{X}$ into **rectangular regions**:

$$f(\mathbf{x}) = \sum_{m=1}^{M} c_m \mathbb{I}(\mathbf{x} \in Q_m),$$

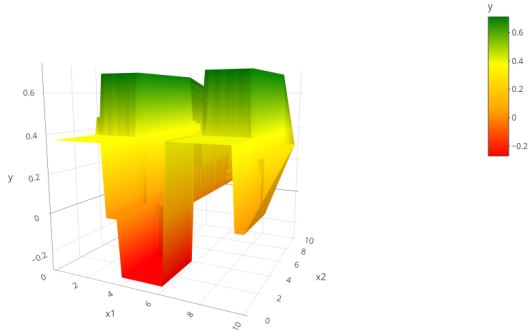
where a tree with $M$ leaf nodes defines $M$ "rectangles" $Q_m$.
$c_m$ is the predicted numerical response, class label or class distribution in the respective leaf node.



# TREE AS AN ADDITIVE MODEL

A 2D regression example:




(For binary classification with probabilities, 2D surface looks similar.)